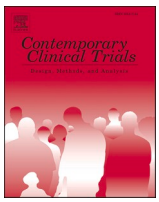
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Psilocybin for alcohol use disorder: Rationale and design considerations for a randomized controlled trial

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ABSTRACT

Several lines of evidence suggest that classic psychedelics (5-HT_{2A} receptor agonists or partial agonists) such as psilocybin might facilitate behavior change in individuals with substance use disorders. We conducted a multi-site, double-blind, randomized controlled trial (RCT) to assess the effects of psilocybin-assisted psychotherapy in alcohol-dependent volunteers. In addition to a structured 12-week psychotherapy platform, participants ($n = 96$) were randomly assigned (1:1) to receive either oral psilocybin or an active placebo (oral diphenhydramine) in each of two dosing sessions (at weeks 4 and 8). Initial doses were 25 mg/70 kg psilocybin or 50 mg diphenhydramine, which could be increased in the second session depending on initial response. The psychotherapy platform combined evidence-based, manualized therapy for alcohol dependence with a supportive context for the dosing sessions. All participants were followed in the RCT through week 36. At the end of the RCT, participants who still met safety criteria were offered an open-label psilocybin session. Data collected at screening, baseline and throughout the study included: demographics, measures of alcohol use, subjective response to psilocybin and diphenhydramine, and safety measures. The primary outcome was the proportion of heavy drinking days during the 32 weeks after the first dosing session (i.e., between week 4 and week 36). Secondary outcomes included safety, additional measures of drinking (e.g., abstinence, drinking days, etc.), craving, self-efficacy, and acute effects. We will also explore moderators and mediators of the primary outcome. The primary outcomes will be published elsewhere. In this paper, we describe the protocol and rationale for our design decisions.

1. Introduction

Modern clinical research on classic psychedelics, most notably psilocybin, has steadily progressed from pilot studies that confirmed their safety and feasibility, through early phase trials that established their potential for clinical efficacy. Of the numerous indications for which these compounds have been investigated, alcohol use disorder (AUD) is among the best-studied. Contemporary research into psilocybin-assisted treatment of AUD is supported by results from trials in the first wave of western psychedelic medicine research. In the 1960's, addiction was a frequent target of investigations into the therapeutic potential of classic psychedelics, a class of hallucinogenic serotonin 2A (5-HT_{2A}) agonists that includes lysergic acid diethylamide (LSD), psilocybin, dimethyltryptamine (DMT), dipropyltryptamine (DPT), and mescaline. A contemporary meta-analysis [1] of six randomized trials of LSD therapy for alcohol dependence [2–7] revealed that participants treated with LSD were nearly twice as likely as those in comparator conditions to demonstrate a reduction in drinking (odds ratio 1.96, 95% confidence interval 1.36–2.84) [1]. Clinical research on classic psychedelics ended in the 1970s, after the Controlled Substances Act placed these compounds into the highly restrictive Schedule I class.

Although early research on the therapeutic utility of classic psychedelics focused on LSD, recent trials in the present wave of psychedelic research have favored psilocybin, due to its excellent safety profile and

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shorter duration of consciousness-altering effects. Dr. Michael Bogenschutz (co-author and principal investigator of the current trial) conducted a small, open-label pilot study of psilocybin-assisted psychotherapy in alcohol-dependent participants (NCT01534494) [8]. In that proof-of-concept study, abstinence increased following treatment with moderately high doses of psilocybin, and the intensity of effects in the medication session predicted changes in craving and self-efficacy in the following weeks [8].

Building on the results of the pilot study, we designed a larger trial investigating the clinical efficacy and psychological mechanisms of psilocybin-assisted psychotherapy for treatment of AUD. Several key decisions were made in the design of this trial based on practical, clinical, and scientific knowledge surrounding the implementation of psilocybin-assisted psychotherapy within the context of a double-blind clinical trial. Here, we present a detailed overview of participant recruitment, selection, and treatment within our recently closed trial of psilocybin-assisted psychotherapy for AUD, and share the rationale behind several strategic protocol decisions. We discuss safety considerations, selection of study drug and choice of comparator, determination of dosing parameters, specification of a psychotherapy platform, and our decision to include an open-label extension phase, and alternative design options and knowledge gaps are explored.

2. Study design

2.1. Study design

Our study (NCT02061293) was a multi-site, parallel-group, 54-week, double-blind randomized controlled trial comparing the effects of two doses of psilocybin versus active placebo (diphenhydramine) on drinking behavior in treatment-seeking individuals who met DSM-IV criteria for alcohol dependence (Fig. 1). During the 12-week treatment phase, the trial was conducted at two sites: the New York University Grossman School of Medicine and the University of New Mexico. Inclusion and exclusion criteria are detailed in Table 1. Eligible participants received a 12-week course of treatment, which included a total of 11 manualized psychotherapy sessions and two full-day (8-h) drug administration sessions, which took place in weeks 4 and 8. The dose of medication administered in the second session was titrated based on response in the first. Assessment visits occurred at set intervals during the treatment phase, and for an additional 24 weeks thereafter. Assessments (Table 2) measured safety, drinking outcomes, and changes in several potential mediators of treatment effect, including motivation, self-efficacy, craving, depression, anxiety, and spiritual dimensions of the experience. The primary outcome measure for the RCT was percent heavy drinking days, measured by timeline follow-back at week 36. At week 38, an optional third, open-label psilocybin session embedded within 6 additional weeks of treatment (4 psychotherapy sessions during weeks 37–42) was available to all participants who completed the double-blind phase of the trial and continued to meet safety criteria. These participants were then followed longitudinally until the final assessment visit at week 54.

2.1.1. Research aims

The primary aim of this trial was to assess the safety and impact of psilocybin (versus diphenhydramine, both administered orally, in conjunction with a psychotherapy platform) on the proportion of heavy drinking days during the 32 weeks after the first dosing session (i.e., between week 4 and week 36). The primary outcomes have been published elsewhere [9]. Secondary objectives, the assessment of which is ongoing, are to characterize the acute effects of psilocybin on alcohol-dependent patients, and to compare psilocybin versus diphenhydramine on additional drinking outcomes. Additional aims are to explore measures of acute hallucinogen effects, and changes in craving, self-efficacy, and motivation as mediators of the primary treatment effect.

3. Study population and recruitment

3.1. Study population and inclusion/exclusion criteria

Enrollment for this trial opened in 2015. The initial randomization target was 100–180 participants, with planned sample size re-estimation to be performed following randomization of 100 participants. However, because of barriers to in-person research visits in the setting of the COVID-19 pandemic, enrollment closed in March 2020, following randomization of 96 participants, including 93 who had received at least one dose of study medication. Participants were treatment-seeking volunteers who met DSM-IV criteria for alcohol dependence, and were willing to participate in a 12-week medication-assisted psychotherapy program targeting their alcohol use. Candidates qualified for the study based on the inclusion and exclusion criteria detailed in Table 1. All work on the study was carried out in accordance with The Code of Ethics of the World Medical Association (Declaration of Helsinki) for experiments involving humans, and was approved by the Institutional Review Board at NYU Langone Health. All research volunteers provided informed consent.

3.2. Recruitment procedures

Potential candidates for the trial were recruited through advertising in local print media, and a significant number were self-referred after learning about the study through Clinicaltrials.gov and/or through media (books, news articles) related to psychedelic research. Additional recruitment efforts, including clinical referrals and flyers posted in clinical settings, were used as needed. Informed consent from all participants was obtained at the time of enrollment.

4. Study intervention

4.1. Psychotherapy platform

After qualifying for the study and completing baseline assessments, participants began the psychotherapy portion of the intervention. We developed a model of preparation, support, and integration (PSI) involving individual preparatory and debriefing sessions to provide a supportive context for the medication sessions. Because the participants were treatment-seeking, ethical considerations mandated the inclusion of evidence-based psychotherapy for all participants. We were also interested in exploring whether psilocybin may enhance the effectiveness of psychotherapy focused on AUD. For both of these reasons, in addition to the PSI component, the psychotherapy manual included a separate, alcohol-focused component. Referred to as Motivational Enhancement and Taking Action (META) therapy, this component included motivational enhancement therapy (MET) as well as more action-oriented cognitive/behavioral therapy content. The approach was modeled on interventions used in multisite trials [10–12], and has been described elsewhere [13]. Fig. 1 illustrates how these two interventions interfaced with each other and with the drug administration sessions.

Each participant was assigned two study therapists: one to take the lead on META sessions, and another to take the lead on PSI sessions. Both therapists were present for the full day on each of the medication sessions. If a participant chose not to participate in the second medication session, they continued with the META sessions as scheduled through Week 12.

4.1.1. Randomization

Randomization was performed as close as possible (usually within 3 days) to the first drug administration session. Randomization was stratified by site and consisted of randomly permuted blocks of varied sizes to ensure relative equality of assignment across treatment groups. The study pharmacist for each site assigned treatment sequentially

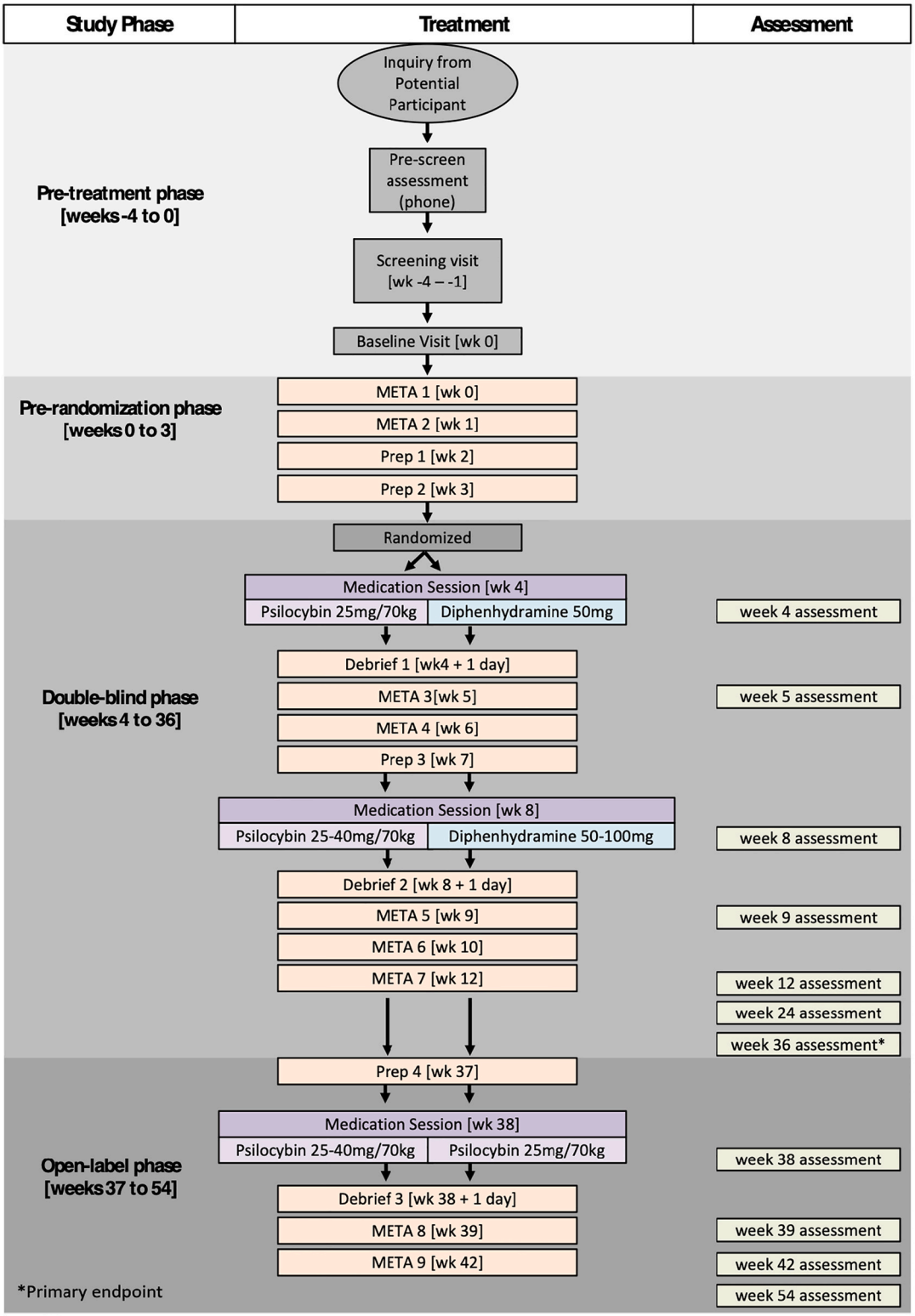


Fig. 1. Participant flow through a clinical trial of psilocybin-assisted psychotherapy for alcohol use disorder. After screening, eligible participants begin a course of psychotherapy (pink boxes) that includes sessions of alcohol-focused Motivational Enhancement and Taking Action (META) therapy and two sessions of Preparation, Support, and Integration therapy (PSI) therapy, a platform designed to support the psychedelic experience. PSI therapy sessions comprise preparatory ("prep") visits and post-dosing integration ("debrief") visits, in addition to the dosing sessions. Shortly before the week 4 visit, participants are randomized to receive either psilocybin or placebo (diphenhydramine) in two dosing sessions, scheduled a month apart (week 4, week 8). Psychotherapy sessions occur before and after the dosing sessions, for a total of twelve active weeks of treatment before the participant enters longitudinal follow-up. The double-blind phase of the trial concludes with the Week 36 assessment. Participants who are interested and eligible are invited to continue in the open-label stage, in which a third dosing session is scheduled for week 38, with additional psychotherapy sessions to support the process. Participants are then followed for additional assessments in weeks 42 and 54. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Inclusion and exclusion criteria.

Inclusion Criteria
Participants must be:

- Males and females age 25–65 with SCID (DSM-IV) diagnosis of alcohol dependence who
- Want to stop or decrease their drinking
- Are not participating in any formal treatment for alcohol dependence (12-step meetings are not considered treatment)
- Are able to provide voluntary informed consent
- Have at least 4 heavy drinking days in the past 30 days (≥4 standard drinks for men, ≥3 for women)
- If female of childbearing potential, are willing to use approved form of contraception[#] from screening until after the psilocybin administration sessions
- Have a family member or friend who can pick them up and stay with them overnight after the psilocybin administration sessions
- Are able to provide adequate locator information.

Exclusion Criteria
Participants are excluded if they have:

- Medical conditions that would preclude safe participation in the trial (e.g., seizure disorder, significantly impaired liver function[*], coronary artery disease, heart failure, uncontrolled hypertension (above 165/95 mmHg at screening)[+], history of cerebrovascular accident, severe asthma[^], hyperthyroidism, narrow-angle glaucoma, stenosing peptic ulcer, pyloroduodenal obstruction, symptomatic prostatic hypertrophy, or bladder-neck obstruction)
- Exclusionary psychiatric conditions (schizophrenia, schizoaffective disorder, bipolar disorder, current major depressive episode, current post-traumatic stress disorder, current suicidality or history of medically serious suicide attempt)
- Cognitive impairment (Folstein Mini Mental State Exam <26 (Folstein, Folstein, & McHugh, 1975)
- A family history of schizophrenia or schizoaffective disorder (first or second degree relatives), or bipolar disorder type 1 (first degree relatives)
- History of hallucinogen use disorder, any use in the past 1 year, or > 25 lifetime uses;
- Cocaine, psychostimulant, opioid, or cannabis dependence (past 12 months)
- Current non-medical use of cocaine, psychostimulants, or opioids (past 30 days)
- Significant alcohol withdrawal (CIWA-Ar score >7. Patients presenting at screening in withdrawal may be referred for detoxification and reassessed within 30 days)
- Serious ECG abnormalities (e.g., evidence of ischemia, myocardial infarction, QTc >0.45 s for men, or > 0.47 s for women)
- Serious abnormalities of complete blood count or chemistries
- Active legal problems with the potential to result in incarceration
- Pregnancy or lactation
- Need to take medication with significant potential to interact with study medications (e.g., antidepressants, antipsychotics, psychostimulants, treatments for addictions, other dopaminergic or serotonergic agents, lithium, anticonvulsants).
- Allergy or hypersensitivity to psilocybin or diphenhydramine.
- High risk of adverse emotional or behavioral reaction based on investigator's clinical evaluation (e.g., evidence of serious personality disorder, antisocial behavior, serious current stressors, lack of meaningful social support).

[#] Acceptable methods of contraception include oral contraceptives, barrier with spermicide, IUD, levonorgestrel implant (Norplant), medroxyprogesterone acetate (Depo provera), surgical sterilization, contraceptive patch, vaginal contraceptive ring, and complete abstinence (not having sex).

[*] Significantly impaired liver function is defined as 1) Alanine aminotransferase (ALT) and/or aspartate aminotransferase (AST) >5 × upper limit of normal (ULN); 2) ALT or AST >3 × ULN with concomitant total bilirubin >2.0 × ULN; or 3) ALT or AST ≥3 × ULN with the appearance of fatigue, nausea, vomiting, right upper quadrant pain or tenderness, fever, rash, and/or eosinophilia.

[+] Participants with elevated blood pressure on screening (i.e., reading above 140/90 mmHg but no higher than 165/95 mmHg) who otherwise qualify for the study may be provisionally included and referred to their primary care provider for management of hypertension. For all participants, vital signs will be measured at all assessment visits, and may be monitored at therapy visits at the discretion of the study physician. Participants receive study medication only if blood pressure is ≤140/90 at safety screening on the day of the drug administration sessions. Participants with a blood pressure reading >165/95 at screening will be excluded from the study.

[^] Severe asthma is defined as asthma diagnosis along with any of the following: 1) an active or chronic cough, 2) current respiratory infection 3) current difficulty breathing 4) asthma-related emergency room visit, hospitalization, or oral, intravenous, or intramuscular steroidal treatment within the past 6 months. Participants with asthma diagnosis that are included in the study and are currently using an inhaler are asked to bring their inhaler to the medication sessions.

according to the randomization code.

4.1.2. Medication sessions

Participants were randomized to receive the same medication (psilocybin or diphenhydramine, delivered orally) in two dosing sessions, which occurred at weeks 4 and 8. Study staff, with the exception of the pharmacist, remained blind to treatment arm and dosing. Participants received both META and PSI sessions prior to and following each medication session, and they were asked to attempt to remain abstinent during the week preceding the two dosing sessions. Dosing sessions were conducted in a dedicated room with supplies and decor elements that deemphasized the clinical setting while promoting a feeling of comfort and safety. Participants were invited to personalize the room (e.g., with personal mementos, sacred objects). Unless otherwise requested by the participant, the room was dimly lit during the session, to reduce external stimulation and promote an inward focus of attention. A set music playlist was selected to further support that inward focus. Starting approximately 30 min after drug ingestion, participants were encouraged to adopt a "default" position of lying supine and wearing eye shades and headphones, while the study therapists continued to hold space and provide support as needed throughout the session.

4.1.3. Psilocybin group

In the first medication session (Week 4), participants randomized to the psilocybin group received psilocybin 25 mg/70 kg. Participants with a BMI exceeding 35 at screening who otherwise qualified for the study received a dose of study medication that corresponded to the body weight that would yield a BMI of 35 for their height. For the second session, the dose was titrated based on response, safety, and participant preference (Fig. 2). If the participant preferred not to increase the dose, they received 25 mg/70 kg again in the second session. If well tolerated, the dose for the second session was the same or higher, based on participant preference, intensity of subjective experience, and QTc (Fig. 2).

4.1.4. Diphenhydramine group

Participants in the placebo group received diphenhydramine 50 mg in the first dosing session. If no clinically significant adverse psychiatric or medical effects were experienced, and if the participant was willing, they received 100 mg for the second session.

4.1.5. Optional extension phase

All participants were followed until at least week 36, the primary endpoint and the end of the double-blind phase of the trial. Participants who completed the double-blind phase and continued to meet safety criteria (including the same parameters for QTc interval, measured on repeat ECG) were eligible to enter the open-label extension phase (see Supplementary Materials).

4.2. Statistical analysis

To enhance transparency and reproducibility of our findings, we composed a formal statistical analysis plan (SAP) developed in accordance with guidelines set forth by Gamble et al. [14]. Analysis samples were defined a priori, and included: the intention to treat (ITT) sample, which comprised all randomized participants; the modified ITT (mITT) sample, which was limited to participants who received study medication at least once; and the per protocol (PP) sample, which was further limited to participants who received study medication and completed minimum psychotherapy requirements (preparation sessions 1 & 2, debrief 1, and META 1–7). To assess safety (Aim 1), blood pressure,

Table 2
Assessment schedule.

Assessments by Visit		Scr.	BL	W04	W05	W08	W09	W12	W24	W36	W38	W39	W42	W54
Medical evaluation	History and physical, ECG, liver function tests, complete blood count, Chem 7, U/A, serum pregnancy test, height and weight, medical history intake form	X												
	Concomitant meds, menstrual calendar	X	X	X	X	X	X	X	X	X	X	X	X	X
	Urine pregnancy test			X		X					X			
Psychiatric assessment	Structured Clinical Interview for DSM-IV (SCID), Family History Questionnaire, Folstein MMSE	X												
Other baseline characteristics	Attentional Resource Allocation Scale, Treatment Goal Form, Demographics Form, Locator Form		X											
Withdrawal	CIWA-Ar	X		X		X					X			
	Time-line Follow-back, BAC	X	X	X		X		X	X	X*	X		X	X
Substance use and consequences	Hair/ Nail sample for ethyl glucuronide								X					
	Urine Drug Screen	X		X		X		X	X	X	X		X	X
	Short Inventory of Problems (past 3 months)		X					X	X	X				X
Acute Hallucinogen Effects	5D-ASC, HRS, SCQ, Monitor Rating Scale, Mysticism scale, guesses of medication received, MUSE questionnaire			X		X					X			
Persisting Effects	Mysticism Scale (lifetime), Spiritual Transcendence Scale, Persisting Effects Questionnaire (follow-up only), BMMRS, NEO-PI-3, Schwartz Value Survey		X							X				X
Motivation	Readiness rulers		X	X	X	X	X	X	X	X	X	X	X	X
Self-efficacy	Alcohol Abstinence Self-efficacy Scale		X	X	X	X	X	X	X	X	X	X	X	X
Craving	Penn Alcohol Craving Scale		X	X	X	X	X	X	X	X	X	X	X	X
Self-Compassion	Self-Compassion Scale (Short Form)		X	X	X	X	X	X	X	X	X	X	X	X
Mood/Anxiety	Ham-D, Ham-A		X	X	X	X	X	X	X	X	X	X	X	X
	Adverse Events Case Report Form			X	X	X	X	X	X	X	X	X	X	X
Safety	Vital Signs and weight	X	X	X	X	X	X	X	X	X	X	X	X	X
	Visual Analog Scales for abuse potential							X	X	X			X	X
Treatment Satisfaction	Visual Analog Scales							X					X	

* Primary outcome.

heart rate, and monitor ratings from item scores were contrasted within the MMRM framework including fixed, categorical effects of treatment and assessment, a treatment-by-assessment interaction, and a fixed covariate (final score on each outcome prior to medication administration). Treatment-emergent adverse events (TEAEs) and number of participants experiencing TEAEs were tabulated, and treatment groups contrasted using Chi squared or Fisher's exact test. Scores from the 5D-ASC and other hallucinogen ratings scales (mean, SD, median, range) were contrasted by group using analysis of variance (at week 4: psilocybin 25 mg/70 kg vs. diphenhydramine 50 mg; at week 8: psilocybin 25 mg/70 kg vs. diphenhydramine 50 mg vs. psilocybin 30 mg/70 kg vs. psilocybin 40 mg/70 kg vs. diphenhydramine 100 mg). Mean scores were compared descriptively to means reported in other studies, particularly Griffiths et al. [15,16]. Double-blind efficacy (Aim 2) was assessed with multivariate repeated measures analysis of variance (RM MANOVA) including fixed, categorical effects of treatment, assessment, and site; site-by-treatment and treatment-by-assessment interactions; fixed covariates for each dependent measure (week 1–4 PHDD, PDD, DPD); and monthly values of PHDD, PDD, and DPD (weeks 5–8, 9–12, 13–16, 17–20, 21–24, 25–28, 29–32, 32–36) as a nested multivariate dependent measure. Significant multivariate treatment effects were decomposed with univariate repeated-measures F tests within each drinking metric (PHDD, PDD, DPD). Methods for open-label efficacy (Aim 5) paralleled those used to evaluate double-blind outcomes, except that only participants in the control group were included in this analysis, with the time effect being of principal interest. Tests of statistical mediation of the treatment effect (Aims 3–4) used a product-of-coefficients approach with a bootstrapping procedure [17] [18].

5. Rationale and design considerations

Psychedelic-assisted psychotherapy opens an exciting new avenue for addiction treatment, but also introduces new challenges and questions surrounding its implementation and evaluation. Our team considered several key issues during the development of the present protocol, identifying a number of unanswered research questions in the process. Safety considerations and our rationale for drug and dosing strategy, comparator condition, psychotherapy framework, and the inclusion of an open-label psilocybin session are discussed in detail below, including advantages and limitations of our approach.

5.1. Safety considerations

We incorporated several safety elements into the design of this trial, including extensive screening procedures, stringent inclusion/exclusion criteria, repeated electrocardiogram (ECG) and blood pressure assessments, ample psychotherapeutic support, dyadic therapy teams, and emergency medications to treat pain, hypertension, or anxious or psychotic reactions. As in most clinical trials, these measures serve to align the study sample with categorical diagnoses, limit the heterogeneity of the sample, exclude certain situations and conditions that might affect outcome independent of any treatment effect, ensure that participants are likely to be able to adhere to the protocol, and maximize safety. Nonetheless, these rigorous safety measures rendered many prospective participants ineligible. Some of these parameters were inflexible due to medical and psychiatric concerns. The medical criteria of most concern are cardiovascular, since psilocybin raises blood pressure and heart rate, and in high doses produces a slight increase in QTc [19]. Some of our safety measures are speculative, and warrant independent investigation to determine whether they can be relaxed in future trials. Personal or family history of psychotic illness was exclusionary because of the possibility that these characteristics may increase the risk of psychotic reactions to psilocybin, although there is no direct evidence for this. Participants were required to have a goal of reducing their drinking, and

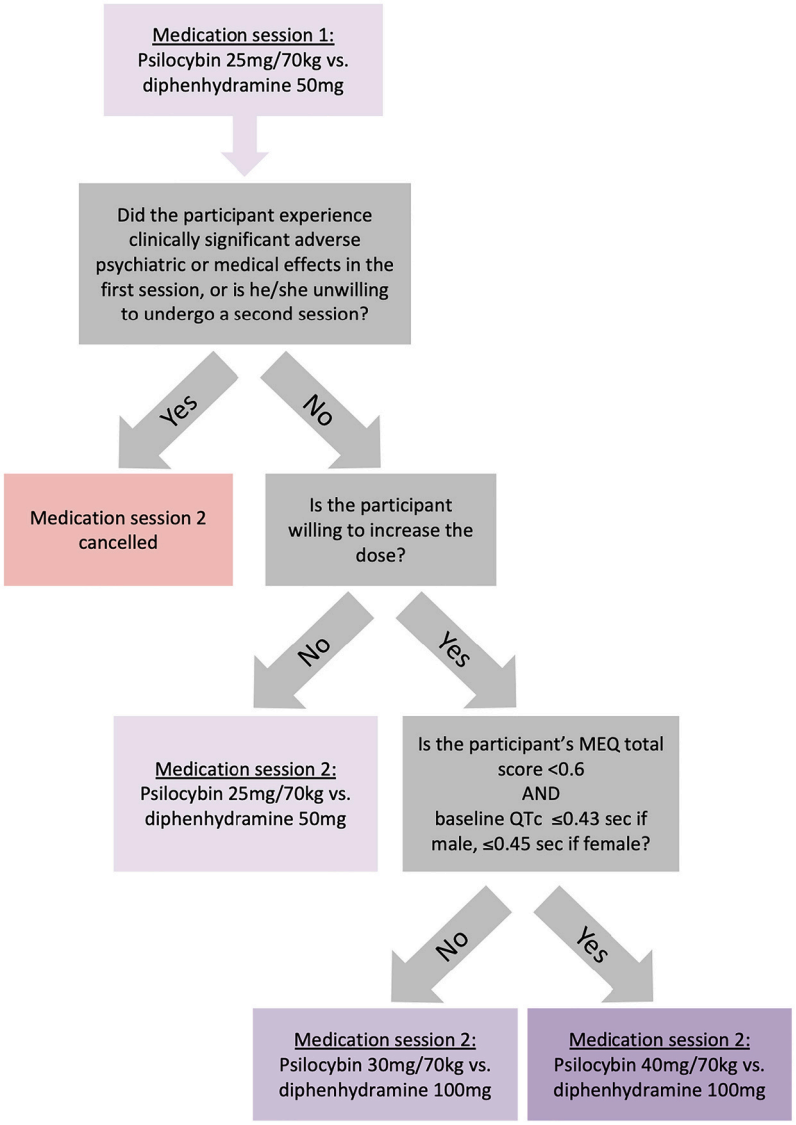


Fig. 2. Algorithm for dosing the study medication. All participants in the double-blind stage of the trial have the opportunity to receive two doses of the study medication, one month apart. The dose of the medication may be raised for the second dosing session. The decision is based on the clinician's impression of medical and psychiatric safety from the first dosing session, the participant's willingness to increase the dose, the QTc as measured by the baseline ECG, and the intensity of the acute effects experienced during the first dosing session, as measured by the Mystical Experiences Questionnaire (MEQ).

to abstain from alcohol for at least 24 h prior to each dosing session. However, a goal of abstinence was not required because there is no evidence that an abstinence goal is necessary to benefit from psilocybin.

5.2. Selection of study drug

Though nearly all of the clinical trials in the first wave of psychedelic research (1950's-1970's) studied the effect of LSD for AUD, we selected psilocybin for this trial and for the pilot study that preceded it. Compared to LSD, which acts for up to twelve hours, the 6-h session typical with psilocybin is more practical for use in the outpatient setting. Moreover, psilocybin is not associated with the same level of cultural stigma as LSD, and recent human studies have established its relatively benign safety profile in appropriately selected participants [16,20–22].

5.3. Psilocybin dosing

The optimal dose of psilocybin in the treatment of alcohol dependence is unknown. Psilocybin is generally well tolerated, but adverse events (e.g., transient anxiety, paranoia, and referential thinking) increase with dose [15,16,23,24]. Individuals vary considerably in their response to a given dosage [8,15,16]. However, there are reasons to believe that high-intensity experiences are more likely to produce clinical benefits. In our pilot study, there was a strong correlation between the total score on the Mystical Experience Questionnaire (MEQ) [16,26] and behavioral outcomes [8]. Clinical trials studying psilocybin-assisted psychotherapy for other indications have found a similar correlation between the MEQ score and clinical outcomes [27,28]. In addition, studies in healthy controls have indicated that a high score on the MEQ is associated with an increase in the personality domain of openness [29] and a lasting sense of personal meaningfulness [30]. We were therefore interested in selecting dosing parameters that optimized the possibility of achieving a strong subjective effect, as measured by the MEQ. There is some evidence from pilot data that alcohol-dependent participants may require higher doses of psilocybin in order to achieve the subjective response that might be obtained in a healthy volunteer [8]. To maximize the probability that participants treated with psilocybin would have at least one strong experience, while minimizing the chance of significant psychological distress or other adverse events, we decided to offer two dosing sessions, with the dose in the second session titrated based on response in the first.

At the time our study was designed, all recent studies of psilocybin had used weight-based dosing, reported either in mg/kg or as the dose that would be administered to a 70 kg participant (e.g., 25 mg/70 kg, which is 0.36 mg/kg), as reported here. While weight-based dosing offers more precise calibration of the drug response for some drugs, there

is no evidence that this is true for psilocybin, which is rapidly converted to its active metabolite, psilocin, and primarily targets the brain. A recent dose-response pharmacokinetics study of doses between 0.3 and 0.6 mg/kg indicated that a fixed dose of 25 mg may approximate 0.3 mg/kg dosing in healthy volunteers [31]. However, higher doses, like those used in the present trial, were not tested in similar simulations. Weight-based dosing requires the presence of a compounding pharmacist, which limits its scalability for future studies. As psychedelic research moves forward, oral formulations will be required to meet the FDA expectations of current Good Manufacturing Practice (cGMP), and the scale of these trials will make weight-based dosing impractical. Some ongoing trials of psilocybin have already begun to use fixed dosing (e.g., 25 mg in a large clinical trial of psilocybin for major depressive disorder, Clinicaltrials.gov ID: NCT03866174). In an analysis of subjective effects reported by 288 participants administered psilocybin in studies conducted at Johns Hopkins University School of Medicine, no significant relationship was found between participant weight and the intensity of subjective effects, regardless of whether dosage was treated as weight-based (20 mg/kg, 30 mg/kg) or approximately fixed (21–25 mg) [32].

5.4. Selection of comparator

There is no consensus as to the best control condition to use in psilocybin trials. It is difficult to create a suitable blind when the active treatment elicits a powerful subjective effect. Methylphenidate, which has been used [16], has known abuse potential, and as such is not ideal in an addiction trial. Niacin, which has been used as a control in some prior trials, causes a highly characteristic discomfort and skin flushing that can unintentionally break the blind. A low dose of psilocybin could be used, but even small dosages might produce some benefits, which could reduce power or obscure the treatment effect. For the active placebo in the present trial, we selected diphenhydramine. There is no evidence that diphenhydramine is effective in the treatment of alcohol dependence. We did not use weight-based dosing for diphenhydramine, as we did not expect it to significantly enhance the blinding. In the dose of 50–100 mg, which is used in this trial, diphenhydramine has low abuse potential [33–36] but noticeable subjective effects, including sedation, confusion, and transient impairments in cognitive and motor functioning [37–40], that might enhance blinding, particularly in participants who are psychedelic-naive.

5.5. Psychotherapy platform

Clinical trials of classic psychedelics have historically been conceptualized as an integration of psychopharmacological and psychosocial interventions—i.e., as studies of psychedelic-assisted psychotherapy. This is in part because appropriate preparation for and integration of the psychedelic experience are understood to be important for mitigating the psychological risks associated with psychedelic use [41]. However, beyond the mitigation of risk, there is limited empirical evidence that a particular psychotherapy framework would enhance the efficacy and/or safety of psychedelics. The PSI model we selected is similar to the supportive framework used in a number of other recent studies of psilocybin [27,28,42]. This model provides a psychosocial package for safe delivery of psilocybin, but includes a non-directive orientation similar to Rogerian client-centered therapy, and allows the participants to select their own goals for the medication and subsequent integration sessions.

Although psychedelic experiences in isolation may cause insights that facilitate change, they do not necessarily determine the content or direction of that change. We therefore chose to supplement the classic PSI platform with a META therapy platform comprising additional therapy sessions aimed at facilitating motivation to change alcohol use behaviors. Though it is not the only evidence-based behavioral intervention for AUD, we selected META therapy because it is well aligned with one of our hypothesized mechanisms for the therapeutic drug effect; to enhance motivation to change. Our META therapy also includes a skills-based cognitive behavioral approach to use if and when a participant moves into the "action" stage of change. The combination of interventions may yield additive or even synergistic effects with psilocybin. Combination models are also being tested in psilocybin-assisted behavioral therapy for tobacco use disorder [43,44] and ketamine with mindfulness training for cocaine use disorder [45]. There is much potential for future studies exploring those issues. Given the high intensity and associated costs of the psychotherapeutic platform, which could limit widespread adoption, there is good reason to explore in future studies whether there is a minimally effective psychotherapy dose.

5.6. Inclusion of an open-label extension phase

Because the subjective effects of psilocybin are so strong, we were concerned about differential drop-out between the two groups, with higher drop-out rates among participants who suspect that they received diphenhydramine in the double-blind medication sessions. To incentivize continued participation for both groups, an open-label extension phase was included after the end of the double-blind phase. In addition to incentivizing continued participation, this extension phase allows the collection of additional safety and tolerability data. There are also downsides to including an open-label session, including limiting the duration of the double-blind follow-up period, limiting the duration of integration and therapy, and substantially expanding the overall timeline and cost of the trial. Though this design element was conceived as a way to maximize retention by minimizing disappointment in the participants who suspected they had received the placebo, other study design options for addressing this issue are available, each with their own set of limitations, as we have discussed elsewhere [46].

6. Discussion

Psilocybin, a classic psychedelic, has effects and mechanisms of action that are entirely distinct from any medications that are currently used in treatment of addiction. Though the question was first raised over six decades ago, ours is the first to use rigorous clinical trial design standards to study whether a classic psychedelic can be used to treat AUD. To date, this is the largest RCT of a classic psychedelic using contemporary clinical trials methodology, with an unusually long follow-up period for trials of AUD; it is the first to combine an established evidence-based behavioral therapy with classic psychedelic administration in the treatment of AUD. It is also the first trial designed to investigate the mechanisms of behavioral change that might be mobilized by psilocybin administration in patients with AUD.

Many clear limitations to this trial have been described in the foregoing discussion of design decisions. A number of other limitations should also be noted. Given the rigorous inclusion/exclusion criteria, many prospective participants were screened out, which may limit generalizability of the results. The weight-based dosing in this study required the presence of a compounding pharmacist, which limits its scalability and is likely to change for future studies. Our robust psychotherapy platform is both a strength and a limitation of the study. In particular, the high number of sessions and the presence of two study therapists at all treatment visits may be difficult to translate into real-world practice, given the expense and training required. Should psychedelic-assisted psychotherapy become an FDA-approved treatment for AUD (or any other clinical indication), it is likely to be provided at specialized psychedelic therapy clinics, staffed by trained psychedelic PSI specialists. In that case, it is conceivable that PSI sessions could be provided in those clinics, while sessions focused on treating AUD specifically could be conducted by a counselor in an addiction treatment center, without the PSI therapist present. Variations in the number of doses and in the psychosocial package for delivery of classic hallucinogens have not been the focus of empirical research, and should be studied in the future; likewise, future studies should investigate

potential mechanisms of action, and strive to identify likely responders. These limitations notwithstanding, we expect that the results of this Phase 2 trial will provide valuable information about the efficacy of psilocybin-assisted psychotherapy for alcohol dependence.

Trial status

NCT02061293 has now closed. The results of primary outcomes have been published elsewhere [9]. Analysis of secondary outcomes is ongoing.

Authors' contributions

All authors contributed substantially to the conception and design of the manuscript and provided important intellectual content. CRediT roles are as follows:

KO: Methodology, investigation, writing - original draft.

SM: Methodology, data curation, investigation, formal analysis, project administration, writing - original draft.

LO: Methodology, investigation, project administration, writing - reviewing and editing.

TB: Methodology, investigation, project administration, writing - reviewing and editing.

SP: Methodology, investigation, project administration, writing - reviewing and editing.

AF: Conceptualization, methodology, investigation, writing - reviewing and editing.

JR: Conceptualization, methodology, writing - reviewing and editing.

SR: Conceptualization, methodology, investigation, writing - reviewing and editing.

MB: Conceptualization, methodology, supervision, funding acquisition, writing - original draft.

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Declaration of Competing Interest

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Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cct.2022.106976.

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